Haitian (Daniel) Zhu

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Education

University of Illinois Chicago Chicago, IL
B.S. in Computer Science Expected May 2027
Honors: Dean's List (Spring 2026) | **Certification:** App Development with Swift Associate (Apple, Dec 2025)
Relevant Coursework: Data Structures, Algorithms, Software Design, Software Engineering, Database Systems, Computer Systems, Programming Language Design & Implementation

Technical Skills

Languages: Swift, TypeScript, JavaScript, Java, Python, SQL, Dart, C, C++
Mobile & Frontend: SwiftUI, Apple Vision, HealthKit, ActivityKit, React, React Native, Expo, Flutter, Leaflet
Backend & Data: Node.js, Fastify, Cloudflare Workers, SQLite (GRDB), IndexedDB (Dexie), Supabase, WebSockets
Testing & Tools: Swift Testing, XCTest, Vitest, Jest, JUnit, Git, GitHub Actions, Xcode, Jira, Claude Code, Codex

Selected Projects

Movement Tracker+ - AI Workout Coach for iPhone Oct 2026 - Nov 2026
Solo, Meta Global AI Developer Hackathon 2026 (Indie Track) | SwiftUI, Vision, HealthKit, Muse API, Cloudflare Workers

- Extended the idea of the CS 440 Workout Movement Tracker (below) into a native iOS coach rebuilt solo in one week; counts reps on-device from Vision body pose, measuring depth, tempo, velocity loss, and partial reps.
- Integrated Meta's Muse model for set reviews, session debriefs, coaching chat, and meal-photo logging through strict JSON schemas; rejected any review citing a nonexistent rep so on-device measurements stay the source of truth.
- Let users teach a new exercise by demonstration, saving it as a counted exercise only when replaying the recording through the engine reproduces the stated rep count.
- Synced workouts, meals, and weight with Apple Health, queuing writes the user has not yet granted; added a Live Activity rest timer, Siri App Shortcuts, and English/Simplified Chinese localization.
- Deployed a Cloudflare Worker proxy that keeps Muse and USDA API keys server-side, with per-install tokens and Durable Object daily budgets; ran 1,300+ Swift and proxy tests in GitHub Actions CI.

OnlyPlanes - Real-Time Flight Tracking App Apr 2026 - Present
Primary Engineer | React Native, Expo, TypeScript, Fastify, SQLite

- Built an iOS/Android flight-tracking app and Fastify backend in a TypeScript monorepo, sharing typed contracts across the mobile app, API, and admin console.
- Delivered viewport-filtered WebSocket updates with timestamp-based position interpolation; coalesced overlapping map requests to limit upstream polling to active viewing regions.
- Implemented quota-aware provider failover and bounded per-client send queues; added Jest regression tests for backpressure, provider failover, and viewport scheduling.
- Ran Claude Code and Codex agents on feature branches merged via PRs with CI; added a hook that blocks destructive shell commands, read-only audit subagents, and cross-model code review that re-verifies fixes with regression tests.

Litter Lifters - Community Litter Reporting Web App Sep 2026 - Present

Team of 4, CS 442 Software Engineering II | React 19, TypeScript, Vite, Leaflet, Dexie (IndexedDB), Zod, Vitest

- Built the report data layer: Zod validation that rejects missing or out-of-range coordinates, a Dexie/IndexedDB repository with seeded reports, and shared React subscriptions that keep the map, report form, and details view in sync.
- Wrote both release scenarios and their acceptance criteria, defining scope boundaries, validation rules, and failure checks for the Release 1 local app and the Release 2 multi-user site.
- Reviewed teammates' pull requests in a Jira-tracked Scrum process and caught bugs before merge; led the map-first responsive UI redesign with a mobile bottom sheet and keyboard focus management.

Workout Movement Tracker (CS 440 Software Engineering I) Jan 2026 - May 2026

Team Lead | Flutter, Dart, SQLite, Gemini API, ML Kit

- Led a 4-person team building a Flutter workout and nutrition tracker with ML Kit pose detection and HealthKit/Health Connect integration.
- Built the SQLite data layer with repositories and versioned schema migrations; integrated USDA FoodData Central food search with 500 ms debounce, stale-response discarding, and SQLite-backed result caching.
- Integrated Gemini coaching for workouts, diet, and goals; cached responses by input fingerprint to skip repeat API calls.

Networked 3-Card Poker Game Nov 2025

Course Project, CS 342 Software Design | Java, JavaFX, TCP Sockets, JUnit

- Built a thread-per-client TCP server for up to 8 JavaFX clients; wrote 17 JUnit tests for hand rankings and payouts.

Experience

Handshake AI Fellowship May 2026 - Oct 2026

AI Output Evaluator (Contract) | Remote

- Evaluated AI-generated images and web interfaces against task-specific rubrics for instruction adherence, visual quality, and functionality.
- Tested interface interactions, documented usability and functional defects, and assigned comparative ratings with evidence-based written feedback.